

Difference Between JPA, Hibernate, and Spring Data JPA

Java Persistence API (JPA)

- JPA is a specification (JSR 338) for object-relational mapping in Java.
- Defines interfaces and annotations for persisting Java objects to relational databases.
- It does not provide an implementation.
- Common implementations include Hibernate, EclipseLink, OpenJPA.

Hibernate

- Hibernate is a concrete ORM tool that implements JPA.
- It can be used directly or through JPA interfaces.
- Offers additional features like caching and native SQL support.
- Requires manual handling of sessions and transactions.

Spring Data JPA

- Spring Data JPA is a Spring module built on top of JPA.
- It doesn't implement JPA itself but uses an underlying provider like Hibernate.
- Provides a higher abstraction layer, simplifying repository creation.
- Automatically handles transaction management and CRUD operations.

Key Differences Summary

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	Implementation	Framework abstraction
Implements JPA	No	Yes	No (uses provider like Hibernate)
Boilerplate Code	Medium	High	Low
Transaction Mgmt	Manual	Manual	Automatic with @Transactional
Repository Support	No	No	Yes, via JpaRepository
Ease of Use	Moderate	Complex	Easy